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METHOD FOR MANUFACTURING GROUP III NITRIDE POWER SEMICONDUCTOR DEVICE USING EPITAXIAL DIES

Abstract

The present invention relates to a method for manufacturing a Group III nitride power semiconductor device using epitaxial dies, by which a high-quality Group III nitride power semiconductor device can be manufacturing by cutting a substrate having epitaxially grown thereon a Group III nitride semiconductor layer so as to form a plurality of epitaxial dies, and then binding the formed plurality of epitaxial dies to a highly heat-dissipating support substrate.

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Background/Summary

TECHNICAL FIELD

[0001] The present invention relates to a method of manufacturing a group III nitride power semiconductor device using epitaxy dies, and more specifically, to a method of manufacturing a group III nitride power semiconductor device using epitaxy dies, which may manufacture a high-quality group III nitride power semiconductor device by cutting a substrate on which a group III nitride semiconductor layer is epitaxially grown to manufacture a plurality of epitaxy dies and then bonding the plurality of manufactured epitaxy dies to a high heat dissipation support substrate.

BACKGROUND ART

[0002] Sapphire growth substrate wafers, on which high-quality epitaxial thin film of group III nitride semiconductors including gallium nitride (GaN) can be grown, have been widely used for development and mass production in the optical device field including light emitting diodes (LEDs) and laser diodes (LDs), and practical verification of epitaxy and chip quality has already been completed. In particular, an epitaxy quality level of a group III nitride semiconductor including gallium nitride (GaN) grown on a sapphire growth substrate has been improved to a level that can satisfy the performance and quality required for next-generation high-frequency or switching devices. In addition, since it is currently possible to supply growth substrate wafers of up to 12 inches, chip dies such as group III nitride power semiconductor-based transistors (a high electron mobility transistor (HEMT), a junction field effect transistor (JFET), and a metal-oxide-semiconductor FET (MOSFET)) and diodes (PN and Schottky) may be manufactured with high cost-effectiveness.

[0003] However, since there is a disadvantage that the thermal conductivity (35 W/mK) of sapphire is much lower than that of silicon (Si), aluminum nitride (AlN), silicon carbide (SiC), etc., the large amount of heat generated when high-output power semiconductors that handle a high voltage or high current are operated may not be easily dissipated, resulting in a fatal weakness in terms of the performance and long-term reliability of group III nitride power semiconductor devices manufactured on the sapphire wafer.

[0004] Currently, among high-output power semiconductor devices made of the group III nitride semiconductors including gallium nitride (GaN), high-frequency devices for a communication device are designed, developed, and manufactured on a silicon carbide (SiC) growth substrate, and switching devices for an electrical/electronic device are designed, developed, and manufactured on a silicon (Si) growth substrate wafer.

[0005] Semi-insulating silicon carbide (SiC) growth substrate wafers have the best heat dissipation performance among materials used as single-crystal semiconductor wafers, and thus have a great advantage in dissipating the large amount of heat generated when an HEMT for a high-frequency communication device is driven, but since the high manufacturing cost of the wafer causes a high-cost issue and the wafer is made of a material that generates tensile stress that causes cracks due to a difference in coefficient of thermal expansion from a gallium nitride (GaN) material, there is a practical difficulty in securing the quality of an epitaxial thin film by increasing a thickness of the gallium nitride (GaN) epitaxial thin film to manufacture power semiconductor devices to have low-density crystal defects.

[0006] In addition, silicon (Si) growth substrate wafers have a great advantage in that 12-inch growth substrate wafers may be supplied, there is compatibility with silicon (Si)-based complementary metal-oxide semiconductor (CMOS) fab processes, and material costs and manufacturing costs for power semiconductors are very inexpensive, but there are problems such as melt-back etching occurring on a silicon (Si) surface in a growth process of a group III nitride epitaxy thin film, severe wafer bowing occurring due to tensile stress caused by large differences in lattice constants (LC) and coefficient of thermal expansion (CTE) from a semiconductor layer, and epitaxy cracks, and in severe cases, wafer breakage, and thus there is a disadvantage in growing the group III nitride epitaxy thin film thickly to have low-density crystal defects.

[0007] Under the above-described background, to improve the group III nitride power semiconductors in terms of performance, quality, and cost in the related art, the group III nitride power semiconductor device has been manufactured by growing a high-quality group III nitride power semiconductor epitaxy thin film structure on a sapphire initial growth substrate to have low-density crystal defects and then transferring the group III nitride power semiconductor epitaxy thin film onto a support substrate wafer having high heat dissipation characteristics through a wafer-level bonding or wafer to wafer (W2W) process.

[0008] However, according to such a conventional technology, typically, since a silicon (Si), silicon carbide (SiC), or aluminum nitride (AlN) final support substrate wafer having high heat dissipation characteristics has a difference in CTE from the sapphire initial growth substrate as large as 2 ppm or more, although it is preferable to perform an annealing process at a predetermined temperature of 300° C. or higher during the W2W bonding process or after the above process is completed, this is realistically impossible.

[0009] Therefore, the conventional W2W bonding process generally introduces an intermediate bonding layer between wafers to bond dielectric materials (SiO_2, spin on glass (SOG), hydrogen silsesquioxane (HSQ), SiN, SiCN, AlN, SiC, diamond, or Al_2O_3), and in this case, strict conditions such as zero foreign substances (particles) on the surfaces of the two wafers, minimizing a total thickness variation (TTV), and ensuring that surface roughness is less than 0.5 nm should be satisfied at the same time, which may not be easily implemented.

DISCLOSURE

Technical Problem

[0010] The present invention is intended to solve the above conventional problems and is directed to providing a method of manufacturing a group III nitride power semiconductor device using epitaxy dies, which may manufacture a high-quality group III nitride power semiconductor device by cutting a substrate on which a group III nitride semiconductor layer is epitaxially grown to manufacture a plurality of epitaxy dies and then bonding the plurality of manufactured epitaxy dies to a high heat dissipation support substrate.

Technical Solution

[0011] According to the present invention, the above object is achieved by a method of manufacturing a group III nitride power semiconductor device using epitaxy dies, which includes a first operation of epitaxially growing a semiconductor layer on a growth substrate, a second operation of cutting the growth substrate on which the semiconductor layer is grown to manufacture a plurality of dies, a third operation of bonding the semiconductor layer of the die to a support substrate through a bonding layer, a fourth operation of separating the growth substrate from the semiconductor layer, and a fifth operation of cutting the support substrate to which the semiconductor layer is bonded to manufacture a power semiconductor device.

[0012] According to the present invention, the above object is achieved by a method of manufacturing a group III nitride power semiconductor device using epitaxy dies, which includes a first operation of epitaxially growing a semiconductor layer on a growth substrate, a second operation of cutting the growth substrate on which the semiconductor layer is grown to manufacture a plurality of dies, a third operation of adhering the semiconductor layer of the die to a temporary substrate through an adhesive layer, a fourth operation of separating the growth substrate from the semiconductor layer, a fifth operation of bonding a surface of the semiconductor layer from which the growth substrate is separated to a support substrate through a bonding layer, a sixth operation of separating the temporary substrate from the adhesive layer, a seventh operation of etching and removing the adhesive layer, and an eighth operation of cutting the support substrate to which the semiconductor layer is bonded to manufacture a power semiconductor device.

[0013] According to the present invention, the above object is achieved by a method of manufacturing a group III nitride power semiconductor device using epitaxy dies, which includes a first operation of epitaxially growing a semiconductor layer on a growth substrate, a second operation of adhering the semiconductor layer to a temporary substrate through an adhesive layer, a third operation of separating the growth substrate from the semiconductor layer, a fourth operation of cutting the temporary substrate to which the semiconductor layer is adhered to manufacture a plurality of dies, a fifth operation of bonding a surface of the semiconductor layer of the die from which the growth substrate is separated to a support substrate through a bonding layer, a sixth operation of separating the temporary substrate from the adhesive layer, a seventh operation of etching and removing the adhesive layer, and an eighth operation of cutting the support substrate to which the semiconductor layer is bonded to manufacture a power semiconductor device.

Advantageous Effects

[0014] According to the present invention, unlike the conventional wafer to wafer (W2W) bonding method, by cutting a substrate on which a group III nitride semiconductor layer is grown epitaxially to manufacture a plurality of epitaxy dies and then bonding these epitaxy dies to a high heat dissipation support substrate (die to wafer (D2W)), it is possible to minimize defects due to a difference in coefficient of thermal expansion (CTE) between a semiconductor layer and a final support substrate, such as wafer bowing, thereby implementing a group III nitride power semiconductor device such as a transistor or diode that has high quality, high thermal conductivity, and high cost-effectiveness.

[0015] In addition, according to the present invention, since a reinforcement layer including a bonding reinforcement layer and a compressive stress layer, which have high-resistance insulation characteristics, can be formed on an upper surface (between a bonding layer and the semiconductor layer) or lower surface (between the bonding layer and a final support substrate) of the bonding layer to effectively block a leakage current to a lower support substrate (or in a horizontal direction), a low-quality and high-resistance gallium nitride (GaN) buffer layer doped with iron (Fe) or carbon (C) is not needed. Therefore, it is possible to secure a high electron mobility transistor (HEMT) active region such as a high-quality gallium nitride (GaN) channel layer and aluminum nitride gallium (AlGaN) barrier layer by omitting the low-quality and high-resistance gallium nitride (GaN) buffer layer, thereby significantly improving the reliability and performance of a power semiconductor device.

[0016] In addition, according to the present invention, since it is not necessary to directly grow a melt-back etching prevention layer and a compressive stress layer that are essential for the growth substrate of the related art, a high-quality aluminum nitride gallium (AlGaN) barrier layer can be grown on a high-quality group III nitride semiconductor layer. In addition, compared to a method of directly growing the melt-back etching prevention layer and the compressive stress layer on the conventional silicon (Si) growth substrate, a low-defect and high-quality group III nitride semiconductor layer can be grown. In addition, since the growth of the melt-back etching prevention layer and the compressive stress layer is omitted, it is possible to implement the group III nitride power semiconductor structure (particularly, an HEMT) having a smaller thickness than before, thereby improving material costs and yield.

[0017] Meanwhile, the effects of the present invention are not limited to the above-described effects, and may include various effects within a range that is apparent to those skilled in the art from the following descriptions.

Description

DESCRIPTION OF DRAWINGS

[0018] FIG. **1** is a flowchart of a method of manufacturing a group III nitride power semiconductor device using epitaxy dies according to a first embodiment of the present invention.

[0019] FIG. **2** shows a process of manufacturing a group III nitride power semiconductor device using epitaxy dies according to the first embodiment of the present invention.

[0020] FIG. **3** shows a reinforcement layer differently disposed on the group III nitride power semiconductor device manufactured according to the first embodiment of the present invention.

[0021] FIG. **4** is a flowchart of a method of manufacturing a group III nitride power semiconductor device using epitaxy dies according to a second embodiment of the present invention.

[0022] FIG. **5** shows a process of manufacturing a group III nitride power semiconductor device using epitaxy dies according to the second embodiment of the present invention.

[0023] FIG. **6** is a flowchart of a method of manufacturing a group III nitride power semiconductor device using epitaxy dies according to a third embodiment of the present invention.

[0024] FIG. **7** shows a process of manufacturing a group III nitride power semiconductor device using epitaxy dies according to the third embodiment of the present invention.

[0025] FIG. **8** shows a reinforcement layer differently disposed on the group III nitride power semiconductor device manufactured according to the second embodiment or third embodiment of the present invention.

MODES OF THE INVENTION

[0026] Hereinafter, some embodiments of the present invention will be described in detail with reference to exemplary drawings. In adding reference numerals to components in each drawing, it should be noted that the same components have the same reference numerals as much as possible even when they are illustrated in different drawings.

[0027] In addition, in describing embodiments of the present invention, detailed descriptions of related known configurations or functions will be omitted when it is determined that the detailed descriptions obscure the understanding of the embodiments of the present invention.

[0028] In addition, terms such as first, second, A, B, (a), and (b) may be used to describe components of the embodiments of the present invention. These terms are only for the purpose of distinguishing one component from another component, and the nature, sequence, order, or the like of the corresponding component is not limited by the terms.

[0029] A method S**100** of manufacturing a group III nitride power semiconductor device using epitaxy dies according to a first embodiment of the present invention will now be described in detail with reference to the accompanying drawings.

[0030] FIG. **1** is a flowchart of a method of manufacturing a group III nitride power semiconductor device using epitaxy dies according to the first embodiment of the present invention, FIG. **2** shows a process of manufacturing a group III nitride power semiconductor device using epitaxy dies according to the first embodiment of the present invention, and FIG. **3** shows a reinforcement layer **160** differently disposed on the group III nitride power semiconductor device manufactured according to the first embodiment of the present invention.

[0031] As shown in FIGS. **1** and **2**, the method S**100** of manufacturing a group III nitride power semiconductor device using epitaxy dies according to the first embodiment of the present invention includes a first operation S**110**, a second operation S**120**, a third operation S**130**, a fourth operation S**140**, and a fifth operation S**150**.

[0032] In the present invention, an initial growth substrate G is an optically transparent and high-temperature heat-resistant substrate through which a laser beam (single wavelength light) may be 100% transmitted (theoretically) without absorption in a laser lift off (LLO) process to be described below, and a material such as sapphire (α-phase Al_2O_3), $ScMgAlO_4$, 4H-SiC, or 6H-SiC is preferably used preferentially. In addition, the initial growth substrate G preferably has a protrusion shape (e.g., a patterned sapphire substrate (PSS)) patterned regularly or irregularly in various dimensions (size and shape) in microscale or nanoscale to minimize crystal defects inside the group III nitride semiconductor thin film grown thereon.

[0033] In addition, a final support substrate **110** is a substrate that supports a buffer layer **121** and a barrier layer **122** after undergoing each operation of the method S**100** of manufacturing a group III nitride power semiconductor device using epitaxy dies according to the first embodiment of the present invention. The final support substrate **110** is preferably provided as a final silicon (Si) support substrate **110** having high heat dissipation performance, and the final silicon (Si) support substrate **110** may be single-crystalline, polycrystalline, or amorphous and may be made of silicon (Si) having a (111) crystal face, a (110) crystal face, or a (100) crystal face. Furthermore, in addition to the above-described silicon (Si), at least one material selected from materials including diamond, silicon carbide (SiC), aluminum nitride (AlN), and sapphire may be included. In particular, the diamond, silicon carbide (SiC), and aluminum nitride (AlN) may be single-crystalline or polycrystalline.

[0034] The first operation S**110** is an operation of epitaxially growing a semiconductor layer **120** on the initial growth substrate G.

[0035] More specifically, the first operation S**110** is an operation of forming a sacrificial layer **130** on the initial growth substrate G and then growing a high-quality group III nitride semiconductor layer **120** (including the group III nitride semiconductor buffer layer **121** and the barrier layer **122**) on the sacrificial layer **130** as a single layer or multiple layers and specifically, is an operation of growing a high-quality buffer layer **121** on the sacrificial layer **130** formed on the initial growth substrate G as a single layer or multiple layers and growing a high-quality barrier layer **122** on the buffer layer **121** as a single layer or multiple layers. Meanwhile, the semiconductor layer **120** of the present invention is not limited to the buffer layer **121** and the barrier layer **122**, and various layers (e.g., a passivation layer and a p-type semiconductor layer for hole injection) for implementing a power semiconductor device structure such as a transistor or a diode may be formed without limitation.

[0036] Here, the sacrificial layer **130** is a layer required for growing a high-quality group III nitride semiconductor layer **120** (including the group III nitride semiconductor buffer layer **121** and the barrier layer **122**), and is made of a material capable of sacrificial separation by thermal-chemical decomposition reactions caused by a laser beam, and for example, the sapphire initial growth substrate G may contain indium gallium nitride (InGaN), gallium nitride (GaN), aluminum gallium nitride (AlGaN), and indium aluminum nitride (InAlN). The sacrificial layer **130** is grown directly on the initial growth substrate G to minimize crystal defects in the semiconductor layer **120** and serves as a buffer.

[0037] In this case, a single layer or multiple layers made of a high-quality group III nitride (GaN, AlGaN, AlN, InAlN), which is a highly electrical resistive insulator, as a layer other than the high-quality buffer layer **121** and the high-quality barrier layer **122**, may be deposited (grown) on the sacrificial layer **130**.

[0038] In addition, the group III nitride semiconductor layer **120**, that is, the group III nitride semiconductor buffer layer **121** and barrier layer **122**, may be made of a single layer or multiple layers of group III nitride semiconductors and may be made of gallium nitride (GaN), aluminum gallium nitride (AlGaN), aluminum nitride (AlN), aluminum gallium nitride/gallium nitride (AlGaN/GaN SLs) with a superlattice structure, aluminum nitride/gallium nitride (AlN/GaN SLs) with a superlattice structure, aluminum gallium nitride/aluminum nitride (AlGaN/AlN SLs) with a superlattice structure, indium gallium nitride (InGaN), indium aluminum nitride (InAlN), gallium nitride/indium aluminum nitride (GaN/InAlN), aluminum scandium nitride (AlScN), gallium nitride/aluminum scandium nitride (GaN/AlScN), etc., which have high-temperature (HT) and high-resistance (HR) characteristics. A critical quality factor for the group III nitride semiconductor layer **120** is reducing the density of fatal crystal defects, that is, threading dislocations (present in a vertical direction with respect to the initial growth substrate G) (≤Low 108/cm2).

[0039] Meanwhile, since a surface of the semiconductor layer **120** (i.e., a surface of the barrier layer **122**) formed on the initial growth substrate G and a surface of the semiconductor layer **120** (i.e., a surface of the buffer layer **121**) subsequently transferred onto the final support substrate **110** are inverted, a total thickness variation (TTV) should be minimized, the surface roughness should be minimized (RMS<1 nm), and a foreign substance (particles) such as an organic substance and a metallic substance should be minimized after growth so that a predetermined preferred surface of the semiconductor layer **120** may be formed, and as a growth process that can achieve the above, all processes using metal organic chemical vapor deposition (MOCVD) and molecular beam epitaxy (MBE) devices are possible, but the growth process is preferably performed through a process with a relatively low growth temperature. For example, in the case of a gallium nitride (GaN) semiconductor layer **120**, a gallium polarity (Ga-polarity) or nitrogen polarity (N-polarity) surface may be selectively controlled according to the surface treatment and growth conditions of the initial growth substrate G. Typically, when the group III nitride semiconductor layer **120** is grown on the sapphire initial growth substrate G wafer in an MOCVD chamber, while the group III nitride semiconductor layer **120** has a surface with a metal (M: Ga, Al, In) polarity with 3 valence electrons, an interface in direct contact with the sapphire initial growth substrate G has a nitrogen polarity with 5 valence electrons.

[0040] The second operation S**120** is an operation of cutting the initial growth substrate G on which the semiconductor layer **120** is grown at preset intervals to manufacture a plurality of epitaxy dies. That is, the epitaxial die in the present invention means that the initial growth substrate G on which the semiconductor layer **120** is epitaxially grown is cut (diced) into a plurality of dies and sorted according to characteristics. Meanwhile, in the second operation S**120**, an epitaxial protective layer may be formed, and some electrodes **170** may be formed in advance.

[0041] Typically, since diamond, a silicon (Si), silicon carbide (SiC), or aluminum nitride (AlN) final support substrate **110** wafer having high heat dissipation characteristics has a difference in CTE from the sapphire initial growth substrate G as large as 2 ppm or more, although it is preferable to perform an annealing process at a predetermined temperature of 300° C. or higher during the W2W bonding process or after the above process is completed, this is realistically impossible. Therefore, the W2W bonding process generally introduces an intermediate bonding layer between wafers to bond dielectric materials (SiO2, spin on glass (SOG), hydrogen silsesquioxane (HSQ), SiN, SiCN, AlN, SiC, diamond, or Al2O3), and in this case, strict conditions such as zero foreign substances (particles) on the surfaces of the two wafers, minimizing a total thickness variation (TTV), and ensuring that surface roughness is less than 0.5 nm should be satisfied at the same time, which may not be easily implemented.

[0042] Therefore, in the present invention, first, the plurality of epitaxy dies may be manufactured by cutting the initial growth substrate G on which the semiconductor layer **120** is grown, and then the plurality of epitaxy dies may each be bonded on the final support substrate **110** wafer having high heat dissipation performance (die to wafer (D2W)), and therefore, the effect of the CTE between the small epitaxy dies and the relatively large final support substrate **110** wafer can be minimized, making it easy to manufacture a high-quality group III nitride power semiconductor device.

[0043] The third operation S**130** is an operation of bonding each of the semiconductor layers **120** of the plurality of epitaxy dies to the final support substrate through a bonding layer **150**. That is, the third operation S**130** is an operation of turning over the plurality of epitaxy dies and pressing and bonding the semiconductor layers **120** to the final support substrate **110** at a temperature of lower than 300° C. For example, the semiconductor layer **120** may be bonded to the final support substrate **110** by forming a first bonding layer on the semiconductor layer **120**, that is, the barrier layer **122**, forming a second bonding layer on the final support substrate **110**, and then bonding the first bonding layer to the second bonding layer to form the bonding layer **150**, and the semiconductor layer **120** may be bonded to the final support substrate **110** after the bonding layer **150** is formed only on the semiconductor layer **120** or the final support substrate **110**.

[0044] Conventionally, epitaxy wafer bowing occurs due to thermo-mechanical induced tensile stress caused by the differences in LC and CTE between the initial growth substrate G and the group III nitride semiconductor, but the epitaxy die bonded to the final support substrate **110** of the present invention may be in a stress-relieved state, thereby minimizing wafer bowing to almost zero. In this case, it is possible to minimize stress by setting a bonding process temperature to about room temperature and performing the process, thereby further minimizing wafer bowing.

[0045] In addition, for the bonding layer **150**, a dielectric material whose properties do not change and which has excellent thermal conductivity in the MOCVD chamber (at a temperature of 1000° C. or higher and in a reducing atmosphere) in which the group III nitride semiconductor is grown may be preferentially selected, and for example, silicon oxide (SiO_2, 0.8 ppm), silicon nitride (SiN_x, 3.8 ppm), silicon carbon nitride (SiCN, 3.8 to 4.8 ppm), aluminum nitride (AlN, 4.6 ppm), silicon carbide (SiC, 4.2 ppm), diamond (1.2 ppm), aluminum oxide (Al_2O_3, 6.8 ppm) may be selected, and furthermore, a flowable oxide (FOx) such as SOG (liquid SiO_2) or HSQ may be selected to improve surface roughness.

[0046] Meanwhile, the reinforcement layer **160** may be formed between the bonding layer **150** and the semiconductor layer **120** or between the bonding layer **150** and the final support substrate **110**.

[0047] In this case, when the reinforcement layer **160** is formed between the bonding layer **150** and the semiconductor layer **120**, as shown in FIG. **2**, the first operation S**110** may include epitaxially growing the semiconductor layer **120** on the initial growth substrate G and then forming the reinforcement layer **160** on the semiconductor layer **120**, the second operation S**120** may include cutting the initial growth substrate G on which the reinforcement layer **160** and the semiconductor layer **120** are formed to manufacture the plurality of epitaxy dies, and the third operation S**130** may include bonding the reinforcement layer **160** of the plurality of epitaxy dies to the final support substrate **110** through the bonding layer **150** (see the top figure of FIG. **3**).

[0048] In addition, although a case where the reinforcement layer **160** is formed between the bonding layer **150** and the final support substrate **110** is not shown, in the third operation S**130**, the semiconductor layers **120** of the plurality of epitaxy dies may be bonded to the reinforcement layer **160** through the bonding layer **150** by forming the reinforcement layer **160** on the final support substrate **110** and then forming the bonding layer **150** on the reinforcement layer **160** (see the middle figure of FIG. **3**).

[0049] Here, more specifically, the reinforcement layer **160** includes a bonding reinforcement layer **161** and a compressive stress layer **162**.

[0050] The bonding reinforcement layer **161** is a layer that is formed in contact with the bonding layer **150** and introduced to increase the bonding strength with the bonding layer **150** when the semiconductor layer **120** is bonded to the final support substrate **110** through the bonding layer **150**, and a material constituting the bonding reinforcement layer **161** may include silicon oxide (SiO_2), silicon nitride (SiN_x), silicon carbon nitride (SiCN), SOG, HSQ, etc.

[0051] The compressive stress layer **162** is a layer that is formed on the bonding reinforcement layer **161** to induce compressive stress and is made of a dielectric material with a value larger than the CTE of the final support substrate **110**, for example, a material that induces compressive stress to relieve tensile stress, such as aluminum nitride (AlN, 4.6 ppm), aluminum nitride oxide (AlNO, 4.6 to 6.8 ppm), aluminum oxide (Al_2O_3, 6.8 ppm), silicon carbide (SiC, 4.8 ppm), silicon carbide nitride (SiCN, 3.8 to 4.8 ppm), gallium nitride (GaN, 5.6 ppm), gallium nitride oxide (GaNO, 5.6 to 6.8 ppm), which serves to guide the quality improvement of products through stress control.

[0052] Meanwhile, as shown in FIG. **3**, in the present invention, in some cases, the reinforcement layer **160** may be disposed between the bonding layer **150** and the semiconductor layer **120** or formed between the bonding layer **150** and the final support substrate **110**, and the reinforcement layer **160** may be formed between the bonding layer **150** and the semiconductor layer **120** and between the bonding layer **150** and the final support substrate **110**. In addition, the bonding reinforcement layer **161** or the compressive stress layer **162** may be omitted from the reinforcement layer **160**, and a case where the reinforcement layer **160** between the bonding layer **150** and the final support substrate **110** is omitted and the bonding layer **150** is directly bonded to the final support substrate **110** may be a structure in which a material having a larger CTE than the final support substrate **110**, such as silicon (Si), is deposited on the bonding layer **150** to induce compressive stress together with a bonding function.

[0053] The fourth operation S**140** is an operation of separating the initial growth substrate G from the buffer layer **121** using an LLO technique. The buffer layer **121** exposed by separating the initial growth substrate G has a nitrogen polar surface (N-polar surface).

[0054] Here, the LLO technique is a technique of separating an epitaxy-grown layer from the initial growth substrate G by irradiating a back surface of the transparent initial growth substrate G with an ultraviolet (UV) laser beam having a uniform optical output and beam profile, and a single wavelength. When the initial growth substrate G is separated, the inside of the semiconductor layer **120** transferred onto the final support substrate **110** is in a state in which the stress is completely relieved and maintains a flat state together with the final support substrate **110**. Then, planarization is achieved by removing the sacrificial layer **130** damaged due to the separation of the initial growth substrate G, contaminated surface residues, a low-quality single crystalline thin film region, etc. through chemical mechanical polishing (CMP), dry etching, etc.

[0055] The fifth operation S**150** is an operation of forming a plurality of electrodes **170** on each epitaxy die according to the power semiconductor structure such as a transistor (three electrodes **170**) or a diode (two electrodes **170**) on the buffer layer **121** of the bonded semiconductor layer **120**, forming a passivation layer **180** as needed, then performing annealing for an ohmic contact or Schottky contact, and then thinning and cutting the final support substrate **110** to manufacture a power semiconductor device, which means a plurality of chip dies or IC dies.

[0056] A group III nitride power semiconductor device manufactured by the method S**100** of manufacturing a group III nitride power semiconductor device using epitaxy dies according to the first embodiment of the present invention may have a structure in which the final support substrate **110**, the bonding layer **150**, the barrier layer **122**, and the buffer layer **121** are sequentially stacked, have a structure in which the reinforcement layer **160** is formed between the bonding layer **150** and the final support substrate **110** or between the bonding layer **150** and the barrier layer **122**, and have a structure in which the plurality of electrodes **170** or the passivation layer **180** is formed on an upper surface of the buffer layer **121**.

[0057] A method S**200** of manufacturing a group III nitride power semiconductor device using epitaxy dies according to a second embodiment of the present invention will now be described in detail with reference to the accompanying drawings.

[0058] FIG. **4** is a flowchart of a method of manufacturing a group III nitride power semiconductor device using epitaxy dies according to a second embodiment of the present invention, FIG. **5** shows a process of manufacturing a group III nitride power semiconductor device using epitaxy dies according to the second embodiment of the present invention, and FIG. **8** shows a reinforcement layer **260** differently disposed on the group III nitride power semiconductor device manufactured according to the second embodiment or third embodiment of the present invention.

[0059] As shown in FIGS. **4** and **5**, the method S**200** of manufacturing a group III nitride power semiconductor device using epitaxy dies according to the second embodiment of the present invention includes a first operation S**210**, a second operation S**220**, a third operation S**230**, a fourth operation S**240**, a fifth operation S**250**, a sixth operation S**260**, a seventh operation S**270**, and an eighth operation S**280**.

[0060] In the present invention, the initial growth substrate G is an optically transparent and high-temperature heat-resistant substrate through which a laser beam (single wavelength light) may be

100% transmitted (theoretically) without absorption in an LLO process to be described below, and a material such as sapphire (α-phase Al_2O_3), $ScMgAlO_4$, 4H-SiC, or 6H-SiC is preferably used. In addition, the initial growth substrate G preferably has a protrusion shape (e.g., a PSS) patterned regularly or irregularly in various dimensions (size and shape) in microscale or nanoscale to minimize crystal defects inside the group III nitride semiconductor thin film grown thereon.
[0061] In addition, an intermediate temporary substrate T has a CTE that is the same as or similar to that of a final support substrate **210** to be described below and at the same time, is made of an optically transparent material through which a laser beam (single wavelength light) may be 100% transmitted (theoretically) without absorption in an LLO process to be described below, and it is preferable that the difference in CTE from the final support substrate **210** does not exceed the maximum of 2 ppm. Sapphire is preferred as a material for the intermediate temporary substrate T, which satisfies the above, and silicon carbide (SiC) or glass whose CTE is adjusted to have a difference of 2 ppm or less from the support substrate **210** may be contained.
[0062] In addition, the final support substrate **210** is a substrate that supports a buffer layer **221** and a barrier layer **222** after undergoing each operation of the method S**200** of manufacturing a group III nitride power semiconductor device using epitaxy dies according to the second embodiment of the present invention. The final support substrate **210** is preferably provided as the silicon (Si) support substrate **210** having high heat dissipation performance, and the silicon (Si) support substrate **210** may be single-crystalline, polycrystalline, or amorphous and may be made of silicon (Si) having a (111) crystal face, a (110) crystal face, or a (100) crystal face. Furthermore, in addition to the above-described silicon (Si), at least one material selected from materials including diamond, silicon carbide (SiC), aluminum nitride (AlN), and sapphire may be included. In particular, the diamond, silicon carbide (SiC), and aluminum nitride (AlN) may be single-crystalline or polycrystalline.
[0063] The first operation S**210** is an operation of epitaxially growing a semiconductor layer **220** on the growth substrate G.
[0064] More specifically, the first operation S**210** is an operation of forming a sacrificial layer **230** on the growth substrate G and then growing a high-quality group III nitride semiconductor layer **220** (including the group III nitride semiconductor buffer layer **221** and barrier layer **222**) on the sacrificial layer **230** as a single layer or multiple layers and specifically, is an operation of growing the high-quality buffer layer **221** on the sacrificial layer **230** formed on a growth substrate G as a single layer or multiple layers and growing a high-quality barrier layer **222** on the buffer layer **221** as a single layer or multiple layers. Meanwhile, the semiconductor layer **220** of the present invention is not limited to the buffer layer **221** and the barrier layer **222**, and various layers (e.g., a passivation layer and a p-type semiconductor layer for hole injection) for implementing a power semiconductor device structure such as a transistor or a diode may be formed without limitation.
[0065] Here, the sacrificial layer **230** is a layer required for growing a high-quality group III nitride semiconductor layer **220** (including the group III nitride semiconductor buffer layer **221** and barrier layer **222**), and is made of a material capable of sacrificial separation by thermal-chemical decomposition reactions caused by a laser beam, and for example, the sapphire initial growth substrate G may contain indium gallium nitride (InGaN), gallium nitride (GaN), aluminum gallium nitride (AlGaN), and indium aluminum nitride (InAlN). The sacrificial layer **230** is grown directly on the initial growth substrate G to minimize crystal defects in the semiconductor layer **220** and serves as a buffer.
[0066] In this case, a single layer or multiple layers made of a high-quality group III nitride (GaN, AlGaN, AlN, InAlN), which is a highly electrical resistive insulator, as a layer other than the high-quality buffer layer **221** and the high-quality barrier layer **222**, may be deposited (grown) on the sacrificial layer **230**.
[0067] In addition, the group III nitride semiconductor layer **220**, that is, the group III nitride semiconductor buffer layer **221** and barrier layer **222**, may be made of a single layer or multiple

layers of group III nitride semiconductors and may be made of gallium nitride (GaN), aluminum gallium nitride (AlGaN), aluminum nitride (AlN), aluminum gallium nitride/gallium nitride (AlGaN/GaN SLs) with a superlattice structure, aluminum nitride/gallium nitride (AlN/GaN SLs) with a superlattice structure, aluminum gallium nitride/aluminum nitride (AlGaN/AlN SLs) with a superlattice structure, indium gallium nitride (InGaN), indium aluminum nitride (InAlN), gallium nitride/indium aluminum nitride (GaN/InAlN), aluminum scandium nitride (AlScN), gallium nitride/aluminum scandium nitride (GaN/AlScN), etc., which have high-temperature (HT) and high-resistance (HR) characteristics. A critical quality factor of the group III nitride semiconductor layer **220** is the reduced density of fatal crystal defects, that is, threading dislocations (present in a vertical direction with respect to the initial growth substrate G) (≤Low 108/cm2).

[0068] Meanwhile, since a surface of the semiconductor layer **220** (i.e., a surface of the barrier layer **222**) formed on the growth substrate G and a surface of the semiconductor layer **220** (i.e., a surface of the buffer layer **221**) subsequently transferred onto the intermediate temporary substrate T are inverted, a total thickness variation (TTV) should be minimized, the surface roughness should be minimized (RMS<1 nm), and a foreign substance (particles) such as an organic substance and a metallic substance should be minimized after growth so that a predetermined preferred surface of the semiconductor layer **220** may be formed, and as a growth process that can achieve the above, all processes using MOCVD and MBE devices are possible, but the growth process is preferably performed through a process with a relatively low growth temperature. For example, in the case of a gallium nitride (GaN) semiconductor layer **220**, a gallium polarity (Ga-polarity) or nitrogen polarity (N-polarity) surface may be selectively controlled according to the surface treatment and growth conditions of the growth substrate G. Typically, when the group III nitride semiconductor layer **220** is grown on the sapphire initial growth substrate G wafer in an MOCVD chamber, while the group III nitride semiconductor layer **120** has a surface with a metal (M: Ga, Al, In) polarity with 3 valence electrons, an interface in direct contact with the sapphire initial growth substrate G has a nitrogen polarity with 5 valence electrons. Meanwhile, in the first operation S**210**, an epitaxial protective layer may be formed, and some electrodes **270** may be formed in advance.

[0069] The second operation S**220** is an operation of cutting the initial growth substrate G on which the semiconductor layer **220** is grown at preset intervals to manufacture a plurality of epitaxy dies. That is, the epitaxial die in the present invention means that the initial growth substrate G on which the semiconductor layer **220** is epitaxially grown is cut (diced) into a plurality of dies and sorted according to characteristics.

[0070] Typically, since silicon (Si), diamond, silicon carbide (SiC), or aluminum nitride (AlN) final support substrate **210** wafer having high heat dissipation conductivity characteristics has a difference in CTE from the sapphire initial growth substrate G as large as 2 ppm or more, although it is preferable to perform an annealing process at a predetermined temperature of 300° C. or higher during the W2W bonding process or after the above process is completed, this is realistically impossible. Therefore, the W2W bonding process generally introduces an intermediate bonding layer **250** between wafers to bond dielectric materials (SiO_2, SOG, HSQ, SiN, SiCN, AlN, SiC, diamond, or Al_2O_3), and in this case, strict conditions such as zero foreign substances (particles) on the surfaces of the two wafers, minimizing a total thickness variation (TTV), and ensuring that surface roughness is less than 0.5 nm should be satisfied at the same time, which may not be easily implemented.

[0071] Therefore, in the present invention, first, the plurality of epitaxy dies may be manufactured by cutting the initial growth substrate G on which the semiconductor layer **220** is grown, and then the plurality of epitaxy dies may each be bonded on the final support substrate **210** wafer having high dissipation performance through the intermediate temporary substrate T (die to wafer (D2W)), and therefore, the effect of the CTE between the small epitaxy dies and the relatively large final support substrate **210** wafer can be minimized, making it easy to manufacture a high-quality group III nitride power semiconductor device.

[0072] The third operation S**230** is an operation of bonding each of the semiconductor layers **220** of the plurality of epitaxy dies to the intermediate temporary substrate T through an adhesive layer **240**. That is, the third operation S**230** is an operation of turning over the plurality of epitaxy dies and pressing and bonding the semiconductor layers **220** to the intermediate temporary substrate T at a temperature of lower than 300° C. For example, the semiconductor layer **220** may be bonded to the intermediate temporary substrate T by forming a first adhesive layer on the semiconductor layer **220**, that is, the barrier layer **222**, forming a second adhesive layer on the intermediate temporary substrate T, and then adhering the first adhesive layer to the second adhesive layer to form the adhesive layer **240**, and the semiconductor layer **220** may be adhered to the intermediate temporary substrate T after the adhesive layer **240** is formed only on the semiconductor layer **220** or the intermediate temporary substrate T.
[0073] In addition, the adhesive layer **240** may contain materials such as silicon oxide (SiO2), SOG, flowable oxide (FOx), silicon nitride (SiNx), aluminum oxide (Al2O3), aluminum nitride (AlN), and silicon carbon nitride (SiCN) as a dielectric material capable of direct bonding at a temperature of 100° C. or lower and contain materials such as resin, benzocyclobutene (BCB), and polyimide (PI) as an organic adhesive capable of indirect bonding at a temperature of 100° C. or lower.
[0074] Meanwhile, the optically transparent intermediate temporary substrate T is a substrate that is finally easily separated by an LLO technique in the subsequent process, and the sacrificial layer **230** may be deposited on the intermediate temporary substrate T before forming the adhesive layer **240**, and a material for the sacrificial layer **230** may contain an oxide, a nitride, etc. that may be deposited by a physical vapor deposition (PVD) technique such as sputtering, pulsed laser deposition (PLD), or an evaporator, specifically, contain a material such as indium tin oxide (ITO), gallium oxide (GaOx), gallium oxide nitride (GaON), gallium nitride (GaN), indium gallium nitride (InGaN), tin oxide (ZnO), indium gallium tin oxide (InGaZnO), indium tin oxide (InZnO), or indium gallium oxide (InGaO),
[0075] The fourth operation S**240** is an operation of separating the initial growth substrate G from the buffer layer **221** using an LLO technique. The buffer layer **221** exposed by separating the initial growth substrate G has a nitrogen polar surface (N-polar surface).
[0076] Here, the LLO technique is a technique of separating an epitaxy-grown layer from the initial growth substrate G by irradiating a back surface of the transparent initial growth substrate G with an ultraviolet (UV) laser beam having a uniform optical output and beam profile, and a single wavelength. When the initial growth substrate G is separated, the inside of the semiconductor layer **220** transferred onto the final support substrate **210** is in a state in which the stress is completely relieved and maintains a flat state together with the final support substrate **210**. Then, planarization is achieved by removing the sacrificial layer **230** damaged due to the separation of the initial growth substrate G, contaminated surface residues, a low-quality single crystalline thin film region, etc. through CMP, dry etching, etc.
[0077] The fifth operation S**250** is an operation of bonding surfaces of the semiconductor layers **220** of the plurality of epitaxy dies from which the initial growth substrate G is separated to the support substrate **210** through the bonding layer **250**. That is, the fifth operation S**250** is an operation of turning over the intermediate temporary substrate T to which the plurality of epitaxy dies are adhered and pressing and bonding the semiconductor layer **220** to the final support substrate **210** at a temperature of lower than 300° C. For example, the semiconductor layer **220** may be bonded to the final support substrate **210** by forming a first bonding layer on the semiconductor layer **220**, that is, the buffer layer **221**, forming a second bonding layer on the final support substrate **210**, and then bonding the first bonding layer to the second bonding layer to form the bonding layer **250**, and the semiconductor layer **220** may be bonded to the final support substrate **210** after the bonding layer **250** is formed only on the semiconductor layer **220** or the final support substrate **210**.

[0078] Conventionally, epitaxy wafer bowing occurs due to thermo-mechanical induced tensile stress caused by the differences in LC and CTE between the initial growth substrate G and the group III nitride semiconductor, but the epitaxy die bonded to the final support substrate **210** through the intermediate temporary substrate T according to the present invention may be in a stress-relieved state, thereby minimizing wafer bowing to almost zero. In this case, it is possible to minimize stress by setting a bonding process temperature to about room temperature and performing the process, thereby further minimizing wafer bowing.
[0079] In addition, for the bonding layer **250**, a dielectric material whose properties do not change and which has excellent thermal conductivity in the MOCVD chamber (at a temperature of 1000° C. or higher and in a reducing atmosphere) in which the group III nitride semiconductor is grown may be preferentially selected, and for example, silicon oxide (SiO_2, 0.8 ppm), silicon nitride (SiN_x, 3.8 ppm), silicon carbon nitride (SiCN, 3.8 to 4.8 ppm), aluminum nitride (AlN, 4.6 ppm), silicon carbide (SiC, 4.2 ppm), diamond (1.2 ppm), aluminum oxide (Al_2O_3, 6.8 ppm) may be selected, and furthermore, a flowable oxide (FOx) such as SOG (liquid SiO_2) or HSQ may be selected to improve surface roughness.
[0080] Meanwhile, the reinforcement layer **260** may be formed between the bonding layer **250** and the semiconductor layer **220** or between the bonding layer **250** and the final support substrate **210**.
[0081] In this case, although a case where the reinforcement layer **260** is formed between the bonding layer **250** and the semiconductor layer **220** is not shown, in the fourth operation S**240** and the fifth operation S**250**, after the initial growth substrate G is separated from the semiconductor layer **220**, the reinforcement layer **260** may be formed on the semiconductor layer **220** (i.e., the buffer layer **221**), and the reinforcement layers **260** of the plurality of epitaxy dies may be bonded to the final support substrate **210** through the bonding layer **250** (see the top figure of FIG. **8**).
[0082] In addition, although a case where the reinforcement layer **260** is formed between the bonding layer **250** and the final support substrate **210** is not shown, in the fifth operation S**250**, the semiconductor layers **220** of the plurality of epitaxy dies may be bonded to the reinforcement layer **260** through the bonding layer **250** by forming the reinforcement layer **260** on the final support substrate **210** and then forming the bonding layer **250** on the reinforcement layer **260** (see the middle figure of FIG. **8**).
[0083] Here, more specifically, the reinforcement layer **260** includes a bonding reinforcement layer **261** and a compressive stress layer **262**, and since the contents of the bonding reinforcement layer **261** and the compressive stress layer **262** are the same as those of the above-described method of manufacturing a group III nitride power semiconductor device using epitaxy dies according to the first embodiment of the present invention, overlapping descriptions thereof will be omitted.
[0084] Meanwhile, as shown in FIG. **8**, in the present invention, in some cases, the reinforcement layer **260** may be disposed between the bonding layer **250** and the semiconductor layer **220** or formed between the bonding layer **250** and the final support substrate **210**, and the reinforcement layer **260** may be formed between the bonding layer **250** and the semiconductor layer **220** and between the bonding layer **250** and the final support substrate **210**. In addition, the bonding reinforcement layer **261** or the compressive stress layer **262** may be omitted from the reinforcement layer **260**, and a case where the reinforcement layer **260** between the bonding layer **250** and the final support substrate **210** is omitted and the bonding layer **250** is directly bonded to the final support substrate **210** may be a structure in which a material having a larger CTE than the final support substrate **210**, such as silicon (Si), is deposited on the bonding layer **250** to induce compressive stress together with a bonding function.
[0085] The sixth operation S**260** is an operation of separating the intermediate temporary substrate T from the sacrificial layer **230** using an LLO technique.
[0086] The seventh operation S**270** is an operation of etching and removing the sacrificial layer **230** and the adhesive layer **240** and etching the bonding layer **250** to set the size of the chip die. As the adhesive layer **240** is etched and removed, the exposed barrier layer **222** has a group 3 metal

polar surface (M-polar surface) such as gallium (Ga).

[0087] The eighth operation S**280** is an operation of forming a plurality of electrodes **270** on each epitaxy die according to the power semiconductor structure such as a transistor (three electrodes **270**) or a diode (two electrodes **270**) on the barrier layer **222** of the bonded semiconductor layer **220**, forming a passivation layer as needed, then performing annealing for an ohmic contact or Schottky contact, and then thinning and cutting the final support substrate **210** to manufacture a power semiconductor device, which means a plurality of chip dies or IC dies.

[0088] A group III nitride power semiconductor device manufactured by the method S**200** of manufacturing a group III nitride power semiconductor device using epitaxy dies according to the second embodiment of the present invention may have a structure in which the final support substrate **210**, the bonding layer **250**, the buffer layer **221**, and the barrier layer **222** are sequentially stacked, have a structure in which the reinforcement layer **260** is formed between the bonding layer **250** and the final support substrate **210** or between the bonding layer **250** and the buffer layer **221**, and have a structure in which the plurality of electrodes **270** or the passivation layer is formed on an upper surface of the barrier layer **222**.

[0089] A method S**300** of manufacturing a group III nitride power semiconductor device using epitaxy dies according to a third embodiment of the present invention will now be described in detail with reference to the accompanying drawings.

[0090] FIG. **6** is a flowchart of a method of manufacturing a group III nitride power semiconductor device using epitaxy dies according to a third embodiment of the present invention, FIG. **7** shows a process of manufacturing a group III nitride power semiconductor device using epitaxy dies according to the third embodiment of the present invention, and FIG. **8** shows a reinforcement layer **360** differently disposed on the group III nitride power semiconductor device manufactured according to the second embodiment or third embodiment of the present invention.

[0091] As shown in FIGS. **6** and **7**, the method S**300** of manufacturing a group III nitride power semiconductor device using epitaxy dies according to the third embodiment of the present invention includes a first operation S**310**, a second operation S**320**, a third operation S**330**, a fourth operation S**340**, a fifth operation S**350**, a sixth operation S**360**, a seventh operation S**370**, and an eighth operation S**380**.

[0092] Since the contents of the initial growth substrate G, the intermediate temporary substrate T, the final support substrate **310**, and the first operation S**310** in the present embodiment are the same as those of the above-described method of manufacturing a group III nitride power semiconductor device using epitaxy dies according to the second embodiment of the present invention, overlapping descriptions thereof will be omitted.

[0093] The second operation S**320** is an operation of adhering a semiconductor layer **320** to the intermediate temporary substrate T through an adhesive layer **340**. That is, the third operation S**330** is an operation of turning over the initial growth substrate G on which the semiconductor layer **320** is grown and pressing and bonding the semiconductor layer **320** to the intermediate temporary substrate T at a temperature of lower than 300° C. For example, the semiconductor layer **320** may be bonded to the intermediate temporary substrate T by forming a first adhesive layer on the semiconductor layer **320**, that is, a barrier layer **322**, forming a second adhesive layer on the intermediate temporary substrate T, and then adhering the first adhesive layer to the second adhesive layer to form the adhesive layer **340**, and the semiconductor layer **320** may be adhered to the intermediate temporary substrate T after the adhesive layer **340** is formed only on the semiconductor layer **320** or the intermediate temporary substrate T.

[0094] Meanwhile, the optically transparent intermediate temporary substrate T is a substrate that is finally easily separated by an LLO technique in the subsequent process, and a sacrificial layer **330** may be deposited on the intermediate temporary substrate T before forming the adhesive layer **340**, and a material for the sacrificial layer **330** may contain an oxide, a nitride, etc. that may be deposited by a physical vapor deposition (PVD) technique such as sputtering, pulsed laser deposition (PLD), or an evaporator, specifically, contain a material such as indium tin oxide (ITO), gallium oxide (GaOx), gallium oxide nitride (GaON), gallium nitride (GaN), indium gallium nitride (InGaN), tin oxide (ZnO), indium gallium tin oxide (InGaZnO), indium tin oxide (InZnO), or indium gallium oxide (InGaO),

[0095] The third operation S**330** is an operation of separating the initial growth substrate G from a buffer layer **321** using an LLO technique. The buffer layer **321** exposed by separating the initial growth substrate G has a nitrogen polar surface (N-polar surface).

[0096] The fourth operation S**340** is an operation of cutting the intermediate temporary substrate T to which the semiconductor layer **320** is adhered at preset intervals to manufacture a plurality of epitaxy dies. That is, the epitaxial die in the present invention means that the intermediate temporary substrate T to which the semiconductor layer **320** is adhered is cut (diced) into a plurality of dies and sorted according to characteristics.

[0097] Typically, since silicon (Si), silicon carbide (SiC), or aluminum nitride (AlN) final support substrate **310** wafer having high heat dissipation characteristics has a difference in CTE from the sapphire initial growth substrate G as large as 2 ppm or more, although it is preferable to perform an annealing process at a predetermined temperature of 300° C. or higher during the W2W bonding process or after the above process is completed, this is realistically impossible. Therefore, the W2W bonding process generally introduces an intermediate bonding layer **350** between wafers to bond dielectric materials (SiO2, SOG, HSQ, SiN, SiCN, AlN, SiC, diamond, or Al2O3), and in this case, strict conditions such as zero foreign substances (particles) on the surfaces of the two wafers, minimizing a total thickness variation (TTV), and ensuring that surface roughness is less than 0.5 nm should be satisfied at the same time, which may not be easily implemented.

[0098] Therefore, in the present invention, first, the plurality of epitaxy dies may be manufactured by cutting the intermediate temporary substrate T to which the semiconductor layer **320** is adhered, and then the plurality of epitaxy dies may be bonded on the final support substrate **310** wafer having high heat dissipation performance (die to wafer (D2W)), and therefore, the effect of the CTE between the small epitaxy dies and the relatively large final support substrate **310** wafer can be minimized, making it easy to manufacture the high-quality group III nitride power semiconductor device.

[0099] The fifth operation S**350** is an operation of bonding surfaces of the semiconductor layers **320** of the plurality of epitaxy dies from which the growth substrate G is separated to the support substrate **310** through the bonding layer **350**.

[0100] Meanwhile, the reinforcement layer **360** may be formed between the bonding layer **350** and the semiconductor layer **320** or between the bonding layer **350** and the final support substrate **310**. Here, more specifically, the reinforcement layer **360** includes a bonding reinforcement layer **361** and a compressive stress layer **362**.

[0101] In this case, although a case where the reinforcement layer **360** is formed between the bonding layer **350** and the semiconductor layer **320** is not shown, in the fourth operation S**340** and the fifth operation S**350**, after the initial growth substrate G is separated from the semiconductor layer **320**, the reinforcement layer **360** may be formed on the semiconductor layer **320** (i.e., the buffer layer **321**), and the reinforcement layers **360** of the plurality of epitaxy dies may be bonded to the final support substrate **310** through the bonding layer **350** (see the top figure of FIG. **8**).

[0102] In addition, although a case where the reinforcement layer **360** is formed between the bonding layer **350** and the final support substrate **310** is not shown, in the fifth operation S**350**, the semiconductor layers **320** of the plurality of epitaxy dies may be bonded to the reinforcement layer **360** through the bonding layer **350** by forming the reinforcement layer **360** on the final support substrate **310** and then forming the bonding layer **350** on the reinforcement layer **360** (see the middle figure of FIG. **8**).

[0103] Since the contents that have not been described in detail in the second operation S**320** to the fifth operation S**350** are the same as those of the above-described method of manufacturing the

group III nitride power semiconductor device using epitaxy dies according to the second embodiment of the present invention, overlapping descriptions thereof will be omitted.
[0104] The sixth operation S**360** is an operation of separating the intermediate temporary substrate T from the sacrificial layer **330** using an LLO technique.
[0105] The seventh operation S**370** is an operation of etching and removing the sacrificial layer **330** and the adhesive layer **340** and etching the bonding layer **350** to set the size of the chip die. As the adhesive layer **340** is removed by being etched, the exposed barrier layer **322** has a group 3 metal polar surface (M-polar surface) such as gallium (Ga).
[0106] The eighth operation S**380** is an operation of forming a plurality of electrodes **370** on each epitaxy die according to the power semiconductor structure such as a transistor (three electrodes **370**) or a diode (two electrodes **370**) on the barrier layer **322** of the bonded semiconductor layer **320**, forming a passivation layer as needed, then performing annealing for an ohmic contact or Schottky contact, and then thinning and cutting the final support substrate **310** to manufacture a power semiconductor device, which means a plurality of chip dies or IC dies.
[0107] A group III nitride power semiconductor device manufactured by the method S**300** of manufacturing a group III nitride power semiconductor device using epitaxy dies according to the third embodiment of the present invention may have a structure in which the final support substrate **310**, the bonding layer **350**, the buffer layer **321**, and the barrier layer **322** are sequentially stacked, have a structure in which the reinforcement layer **360** is formed between the bonding layer **350** and the final support substrate **310** or between the bonding layer **350** and the buffer layer **321**, and have a structure in which the plurality of electrodes **370** or the passivation layer is formed on an upper surface of the barrier layer **322**.
[0108] As described above, although all the components constituting embodiments disclosed herein were described as being combined or combined to operate as one, the present invention is not necessarily limited to these embodiments. That is, one or more of all the components may be combined to operate as one without departing from the scope of the purpose of the present invention.
[0109] In addition, the terms such as “comprise,” “constitute,” or “have” described above mean that the corresponding component may be inherent unless otherwise stated, and thus should be construed as further including another component rather than excluding another component. All terms including technical or scientific terms have the same meaning as commonly understood by those skilled in the art to which the present invention pertains unless defined otherwise. Commonly used terms, such as terms defined in a dictionary, should be intended as being consistent with the contextual meaning of the related art and are not interpreted in an ideal or excessively formal meaning unless explicitly defined herein.
[0110] In addition, the above description is merely the exemplary description of the technical spirit of the present invention, and those skilled in the art to which the present invention pertains will be able to variously modify and change the present invention without departing from the essential characteristics of the present invention.
[0111] Therefore, the embodiments disclosed in the present invention are not intended to limit the technical spirit of the present invention, but intended to describe the same, and the scope of the technical spirit of the present invention is not limited by these embodiments. The scope of the present invention should be construed by the appended claims, and all technical ideas within the equivalent scope should be construed as being included in the scope of the present invention.

Claims

1. A method of manufacturing a group III nitride power semiconductor device using epitaxy dies, the method comprising: a first operation of epitaxially growing a semiconductor layer on a growth substrate; a second operation of cutting the growth substrate on which the semiconductor layer is

grown to manufacture a plurality of dies; a third operation of bonding the semiconductor layer of the die to a support substrate through a bonding layer; a fourth operation of separating the growth substrate from the semiconductor layer; and a fifth operation of cutting the support substrate to which the semiconductor layer is bonded to manufacture a power semiconductor device.
2. The method of claim 1, wherein a reinforcement layer, which increases bonding strength and induces compressive stress, is formed between the bonding layer and the semiconductor layer or between the bonding layer and the support substrate.
3. The method of claim 2, wherein the reinforcement layer includes a bonding reinforcement layer formed in contact with the bonding layer to increase bonding strength with the bonding layer and a compressive stress layer formed on the bonding reinforcement layer to induce compressive stress.
4. The method of claim 1, wherein the fifth operation includes forming a plurality of electrodes on the semiconductor layer.
5. The method of claim 1, wherein the fifth operation includes forming a passivation layer on the semiconductor layer.
6. The method of claim 1, wherein the semiconductor layer includes a buffer layer grown on the growth substrate and a barrier layer grown on the buffer layer.
7. A method of manufacturing a group III nitride power semiconductor device using epitaxy dies, the method comprising: a first operation of epitaxially growing a semiconductor layer on a growth substrate; a second operation of cutting the growth substrate on which the semiconductor layer is grown to manufacture a plurality of dies; a third operation of adhering the semiconductor layer of the die to a temporary substrate through an adhesive layer; a fourth operation of separating the growth substrate from the semiconductor layer; a fifth operation of bonding a surface of the semiconductor layer from which the growth substrate is separated to a support substrate through a bonding layer; a sixth operation of separating the temporary substrate from the adhesive layer; a seventh operation of etching and removing the adhesive layer; and an eighth operation of cutting the support substrate to which the semiconductor layer is bonded to manufacture a power semiconductor device.
8. The method of claim 7, wherein a reinforcement layer, which increases bonding strength and induces compressive stress, is formed between the bonding layer and the semiconductor layer or between the bonding layer and the support substrate.
9. The method of claim 8, wherein the reinforcement layer includes a bonding reinforcement layer formed in contact with the bonding layer to increase bonding strength with the bonding layer and a compressive stress layer formed on the bonding reinforcement layer to induce compressive stress.
10. The method of claim 7, wherein the eighth operation includes forming a plurality of electrodes on the semiconductor layer.
11. The method of claim 7, wherein the eighth operation includes forming a passivation layer on the semiconductor layer.
12. The method of claim 7, wherein the semiconductor layer includes a buffer layer grown on the growth substrate and a barrier layer grown on the buffer layer.
13. A method of manufacturing a group III nitride power semiconductor device using epitaxy dies, the method comprising: a first operation of epitaxially growing a semiconductor layer on a growth substrate; a second operation of adhering the semiconductor layer to a temporary substrate through an adhesive layer; a third operation of separating the growth substrate from the semiconductor layer; a fourth operation of cutting the temporary substrate to which the semiconductor layer is adhered to manufacture a plurality of dies; a fifth operation of bonding a surface of the semiconductor layer of the die from which the growth substrate is separated to a support substrate through a bonding layer; a sixth operation of separating the temporary substrate from the adhesive layer; a seventh operation of etching and removing the adhesive layer; and an eighth operation of cutting the support substrate to which the semiconductor layer is bonded to manufacture a power semiconductor device.

14. The method of claim 13, wherein a reinforcement layer, which increases bonding strength and induces compressive stress, is formed between the semiconductor layer and the bonding layer.
15. The method of claim 14, wherein the reinforcement layer includes a bonding reinforcement layer formed in contact with the bonding layer to increase bonding strength with the bonding layer and a compressive stress layer formed on the bonding reinforcement layer to induce compressive stress.
16. The method of claim 13, wherein the eighth operation includes forming a plurality of electrodes on the semiconductor layer.
17. The method of claim 13, wherein the eighth operation includes forming a passivation layer on the semiconductor layer.
18. The method of claim 13, wherein the semiconductor layer includes a buffer layer grown on the growth substrate and a barrier layer grown on the buffer layer.